



Review Pulse v2 — Aspect-Based Sentiment Analysis

"One review, many opinions — sentiment for each aspect, with attention you can see."

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Approach · Contribution

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Intro · Literature

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Problem · Plan & Risks



THE PROBLEM

Why one label is not enough

"The food was great but the service was slow." One review, two opposite opinions — sentence-level models collapse it to a single neutral.

**One review**

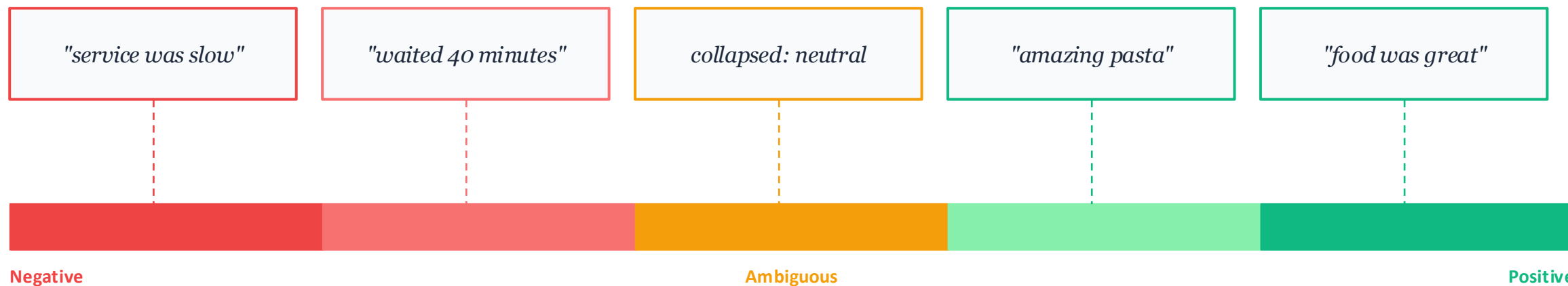
two opposite opinions, one per aspect

**Sentence-level models**

our A1 N-gram and Zhao et al. CNN both collapse to one label

**Business value**

"food good, service bad" is actionable — "neutral" is not

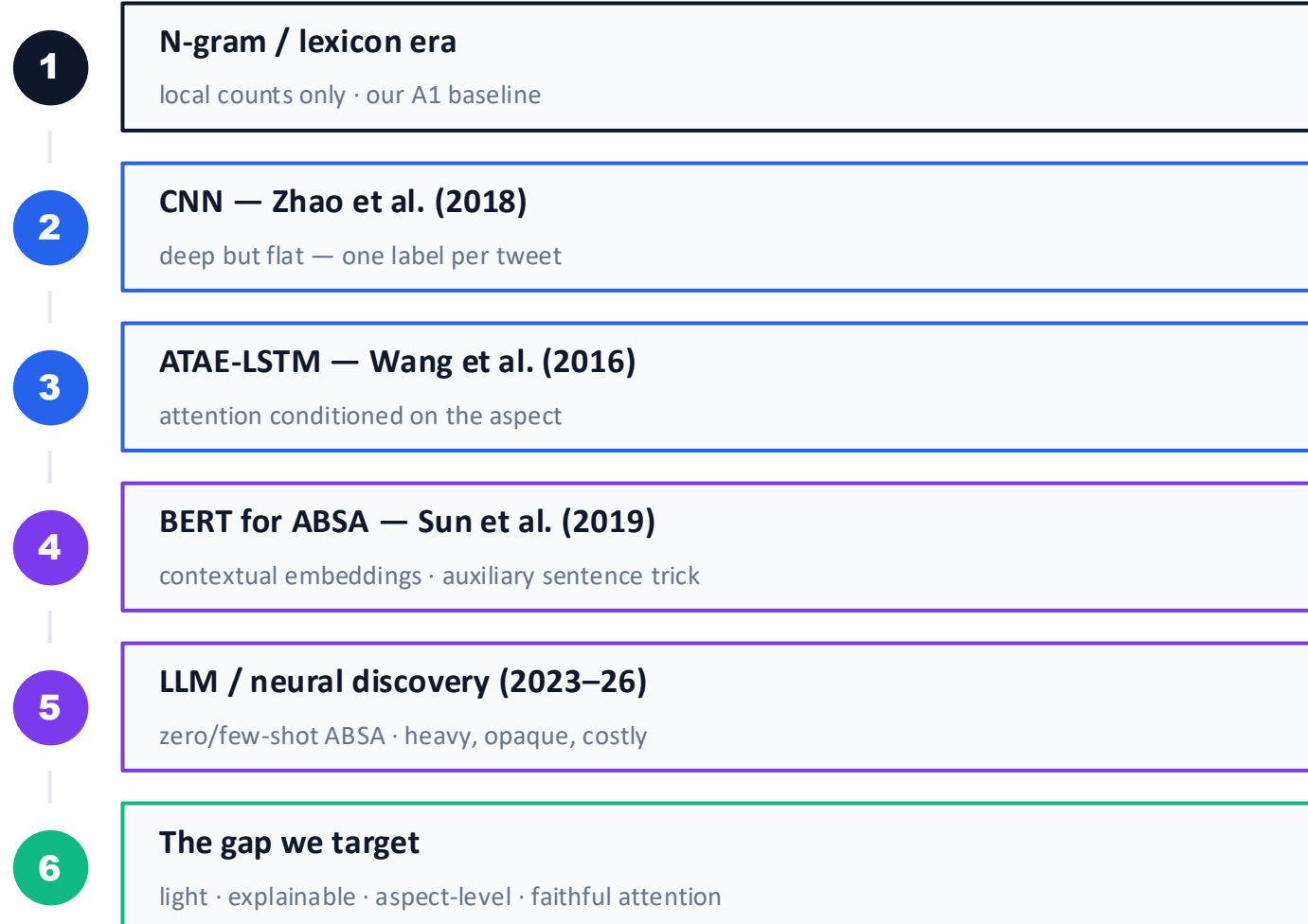


Sentence-level collapse — opposite opinions average into a meaningless neutral



WHAT THE LITERATURE SHOWS

One limitation — a single label per text — solved step by step



THE GAP

Frontier is heavy

THEM

LLM-driven · opaque



US

light · faithful attention

Why it matters

Few works pair a light, explainable model with faithful attention at the aspect level. That is exactly where we sit.



OUR APPROACH

Two models compared — interpretability is the differentiator

MODEL 1 · ATAE-LSTM

Aspect-aware attention

```
Input(
  review + aspect
  GloVe embed 100d
  LSTM hidden 256
)
Attention(
  aspect-conditioned weights
)
```

ROLE

Explainable core · fallback path

STRENGTH

Faithful attention heatmaps

MODEL 2 · TRANSFORMER

DistilBERT fine-tuned

(review, aspect)
(review, aspect) pair input

Per-aspect sentiment
accuracy · macro-F1

Attention weights
per-aspect logits

3-class output
neg · neutral · pos

WHY

Contextual power at low compute — fine-tuned on a free Colab GPU.

DIFFERENTIATOR · XAI

Attention heatmap

AutoTokenizer
SemEval-2014 benchmark

DistilBERT base
66M params · uncased

Classifier head
which words drove each call

SCOPE

1. Restaurants + Laptops — pre-annotated, no manual labelling
2. Topic modelling for aspect discovery — gated stretch goal

We show why, not just the label — per-aspect and transparent.

Modules 8–12: feasible and de-risked

Timeline: data + baseline → ATAE-LSTM → DistilBERT → interpretability + Streamlit demo → evaluation + report.



Small data → overfitting

RISK

Mitigated: dropout · early stopping · transfer learning from pretrained embeddings.



Heavy fine-tuning

RISK

Mitigated: DistilBERT fine-tuned on a free Colab GPU — zero-cost budget.



Scope creep

RISK

Mitigated: aspect extraction & topic modelling are gated stretch goals.



Pre-annotated data

WIN

SemEval-2014 ships with aspect labels — the feasibility win: no manual labelling.



Light compute

WIN

Free Colab · open datasets · open-source libraries — zero-cost end to end.



Working fallback

WIN

If the transformer underperforms, the ATAE-LSTM path still meets the rubric.

EXPECTED CONTRIBUTION

From one score per review to a transparent, per-aspect read



PROPOSED Review Pulse v2 — built and demoed live in Assessment 3

1

ASPECT-LEVEL

Sentiment per thing

Per-aspect labels instead of one collapsed score per review.

2

EXPLAINABLE

Attention heatmap

Shows which words drove each per-aspect call — the why, not just the label. Our differentiator.

3

COMPARED

LSTM vs DistilBERT

Aspect-aware attention-LSTM vs fine-tuned transformer, evaluated on identical benchmarks.

4

LIVE DEMO

Streamlit app

Working demo in Assessment 3 — type a review, watch per-aspect sentiment and attention live.



Questions?

2

SemEval datasets

2

models compared

5

modules · 8 to 12

1

live demo in A3

References: Wang et al. (2016) · Sun et al. (2019) · Devlin et al. (2019) · He et al. (2017) · Pontiki et al. (2014) · Hua et al. (2024) · Zhao et al. (2018) — full APA list in the written proposal.